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Se consideră funcția $f_m(x) = x^{2m+1} + 3x^3 - 1$, $m \in \mathbb{N}^*$

Dacă x_i , $i = \overline{1, 2m+1}$ sunt rădăcinile ecuației $f_m(x) = 0$

Să se calculeze

a) $[x_1, x_2, \dots, x_{2m+1}; x^{2m+h}]$ pentru $h = \overline{2, 2m+1}$

Vom folosi

$$[x_0, x_1, \dots, x_m; x^p] = \begin{cases} 0, & p < m \\ m!, & p = m \\ \sum_{i=0}^m x_i^p, & p = m+1 \end{cases}$$

și

$$[x_0, x_1, \dots, x_m; f] = \sum_{i=0}^m \frac{f(x_i)}{l'(x_i)}$$

unde

$$l(x) = (x - x_0) \dots (x - x_m)$$

$$\left(\sum_{i=0}^m x_i \right)^2 = \sum_{1 \leq i < j \leq m} x_i x_j$$

$$[x_1, x_2, \dots, x_{2m+1}; x^{2m+h}] =$$

$$= [x_1, x_2, \dots, x_{2m+1}; x^{h-1} (x^{2m+1} + 3x^3 - 1) - 3x^{h+2} + x^{h+1}]$$

$$= [x_1, x_2, \dots, x_{2m+1}; x^{h-1} (x^{2m+1} + 3x^3 - 1)] -$$

$$- 3[x_1, x_2, \dots, x_{2m+1}; x^{h+2}] + [x_1, x_2, \dots, x_{2m+1}; x^{h+1}]$$

Avem

$$[x_1, \dots, x_{2m+1}; x^{h-1} (x^{2m+1} + 3x^3 - 1)] = \sum_{i=1}^{2m+1} \frac{x_i^{h-1} (x_i^{2m+1} + 3x_i^3 - 1)}{l'(x_i)} \stackrel{=0}{=}$$

$$\text{unde } l(x) = (x - x_1) \dots (x - x_{2m+1})$$

$$h = \overline{2, 2m+1} \Rightarrow h-1 = \overline{1, 2m} \Rightarrow [x_1, x_2, \dots, x_{2m+1}; x^{h-1}] = 0$$

$$\Rightarrow [x_1, x_2, \dots, x_{2m+1}; x^{2m+h}] = 3[x_1, x_2, \dots, x_{2m+1}; x^{h+2}]$$

$$h = \overline{2, 2m-1} \Rightarrow h+2 = \overline{4, 2m+1}$$

$$= 22$$

$$\Rightarrow [x_1, x_2, \dots, x_{2m+1}; x^{h+1}] = \begin{cases} 1, & h = 2m-1 \\ 0, & h < 2m-1 \end{cases}$$

$$\Rightarrow [x_1, x_2, \dots, x_{2m+1}; x^{2m+h}] = \begin{cases} -3, & h = 2m-1 \\ 0, & h < 2m-1 \end{cases}$$

b) Să se calculeze $[1, \frac{1}{2}, \dots, \frac{1}{10}; \frac{x}{x^2-4}]$

Vom folosi rezultatul

$$[x_0, x_1, \dots, x_m; \frac{1}{ax+b}] = (-1)^m \frac{a^m}{\prod_{i=0}^m (ax_i + b)}$$

$$\frac{x}{x^2-4} = \frac{1}{2} \left(\frac{1}{x-2} + \frac{1}{x+2} \right)$$

$$[1, \frac{1}{2}, \dots, \frac{1}{10}; \frac{x}{x^2-4}] = \frac{1}{2} [1, \frac{1}{2}, \dots, \frac{1}{10}; \frac{1}{x-2}] + \frac{1}{2} [1, \frac{1}{2}, \dots, \frac{1}{10}; \frac{1}{x+2}]$$

$$[x_0, x_1, \dots, x_m; \alpha f + \beta g] = \alpha [x_0, x_1, \dots, x_m; f] + \beta [x_0, x_1, \dots, x_m; g]$$

$$[1, \frac{1}{2}, \dots, \frac{1}{10}; \frac{1}{x+2}] = (-1)^9 \frac{1}{\prod_{i=0}^9 \left(\frac{1}{i+1} + 2 \right)}$$

$x_0=1, x_1=\frac{1}{2}, \dots, x_9=\frac{1}{10} \quad (n=9)$

$$[1, \frac{1}{2}, \dots, \frac{1}{10}; \frac{1}{x-2}] = (-1)^9 \frac{1}{\prod_{i=0}^9 \left(\frac{1}{i+1} - 2 \right)}$$

$$\Rightarrow [1, \frac{1}{2}, \dots, \frac{1}{10}; \frac{x}{x^2-4}] = \frac{1}{2} \frac{1}{\prod_{i=0}^9 \frac{2i+3}{i+1}} + \frac{1}{2} \frac{1}{\prod_{i=0}^9 \frac{-2i-1}{i+1}}$$

$$= \frac{1}{2} \prod_{i=0}^9 (i+1) \left(\frac{1}{\prod_{i=0}^9 (2i+3)} - \frac{1}{\prod_{i=0}^9 (2i+1)} \right) = \frac{10!}{2} \cdot \frac{20}{21!!} = \frac{10 \cdot 10!}{21!!}$$

unde $21!! = 1 \cdot 3 \cdot 5 \cdot \dots \cdot 21$